

EVOLUTION

Genomic architecture of egg mimicry and its consequences for speciation in parasitic cuckoos

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Host-parasite arms races facilitate rapid evolution and can fuel speciation. *Cuculus* cuckoos are deceptive egg mimics that exhibit a broad diversity of counterfeit egg phenotypes, representing host-adapted subpopulations (gentes). Genome analysis of 298 common (*Cuculus canorus*) and 50 oriental cuckoos (*Cuculus optatus*) spanning 15 egg morphs revealed that eggshell background coloration is predominantly influenced by matrilineal genetic variation. Recurrent mitochondrial mutations and an ancient W chromosome-linked translocation of an autosomal assembly factor for respiratory complex I provide a tentative link between mitochondrial function and pigment synthesis through the heme pathway. Biparentally inherited loci contribute to phenotypic variation in both species, mainly for maculation. The evolutionary tug-of-war over a sex-limited, mimetic trait integrates autosomal components with the nonrecombining, matrilineal genome without catalyzing genome-wide divergence between gentes.

The common cuckoo (*Cuculus canorus*) is a textbook example of an obligate brood parasite that has fascinated naturalists since the birth of zoology (1, 2). It exploits more than 100 host species (3, 4) and displays large variation in mimetic egg phenotypes. It has been suggested that host-matched phenotypes evolved in subpopulations specialized in parasitizing a particular host species while maintaining evolutionary independence (host races or gentes) (5, 6). Gens diversification is believed to be fueled by concurrent coevolutionary arms races between cuckoos and their passerine hosts, many of which are capable of rejecting parasitic eggs that are dissimilar from their own (3, 7). Yet for the parasite, the expected evolutionary outcome strongly depends on the genetic architecture and inheritance of the traits involved (8, 9).

Host switches can lead to reproductive isolation and promote speciation if the mimetic trait encompasses both sexes and is encoded on the autosomal genome and if mating is assortative by host (10, 11). This is exemplified in indigobirds, whydahs, and bronze cuckoos, where host mimicry involves nestling traits expressed by both sexes (12, 13). In combination with host-specific assortative mating mediated by

imprinting [e.g., on host song (14)], coevolution has facilitated rapid diversification of parasitic lineages (13, 15). Alternatively, female-limited parasitic traits, if encoded on the matrilineal component of the genome (mtDNA, W chromosome), can be stably inherited from mothers to daughters irrespective of assortative mating (6, 16). Such a genetic architecture is less conducive to speciation (8) and has been suggested for honeyguides (17), cuckoo finches (18), and in small-scale genetic studies for common cuckoos (3, 19–22).

Here, we revisit the two competing hypotheses governing diversification of host races in two brood parasitic *Cuculus* species: the common cuckoo (*C. canorus*; *canorus* hereafter) and the closely related oriental cuckoo (*C. optatus*; *optatus* hereafter) (23). We assess in both species whether host races, operationally defined by egg phenotype, are predicated on matrilineal variation (3, 19–22) or whether biparental inheritance contributes to egg diversity (24, 25).

Geographic structure of genetic variation

We characterized both matrilineal (mtDNA, W chromosome) and coinherited (autosomes, Z chromosome) genetic variation across the distributional range of 202 unrelated *canorus* and 50 *optatus* representing a total of 25 host species (21 *canorus*, 4 *optatus*) and 15 egg morphs (11 *canorus*, 4 *optatus*) (fig. S1 and table S1). Matrilineal phylogenies indicate seven primary haplogroups in *canorus* (M_{CC1} to M_{CC7}) and four in *optatus* (M_{CO1} to M_{CO4}) (Fig. 1A and fig. S2). An ancient split 1.4 million to 2.9 million years ago, exceeding *optatus* species boundaries [<1 million years (23)], separates east Asian *canorus* from European *canorus* (minimum and maximum 95% highest posterior density; figs. S3 to S5). An exception to this East-West geographic divide is the *canorus* lineage of blue egg-laying cuckoos in Finland (M_{CC1}) clustering with haplogroups M_{CC2} and M_{CC3} predominantly found in East Asia (19). Among the remaining four haplogroups (M_{CC4} to M_{CC7}) of the *canorus* lineage, M_{CC4} is closely related to *optatus* lineages M_{CO1} and M_{CO2}. These haplogroups correspond exclusively to females of the rufous plumage morph, which, similar to egg phenotypic variation, appears to be subject to balancing selection (23) (Fig. 1A and figs. S2 and S3). In *canorus*, haplogroups are broadly spread across the species range, except for East Asian M_{CC3} and the blue egg-associated M_{CC1} of Finnish *canorus* (Fig. 1C). In *optatus*, M_{CO1} is maintained across the distributional range, whereas the other haplogroups are geographically restricted (Fig. 1D and fig. S6).

Autosomal population structure in *canorus* comprises five major genetic clusters [A_{CC1} to A_{CC5}, maximum fixation index (F_{ST}) = 0.085], each aligning with broad geographic areas across Eurasia (Fig. 1B and figs. S7 and S8). At finer resolution among 14 geographically defined groups (geographic clusters G_{CC1} to G_{CC14}; fig. S9), significant isolation by distance is evident (Spearman's ρ = 0.92, p = 2.4×10^{-18}). By contrast, matrilineal variation shows much weaker isolation by distance (W chromosome ρ = 0.33, p = 0.04) (Fig. 1, C and E, and fig. S10), counter to expectations from reduced effective population size but consistent with female-biased dispersal (26, 27). These results highlight an association primarily between geographic location and biparental autosomal variation (Fig. 1E). Meanwhile, deeply divergent haplogroups are widely dispersed across the *canorus* range, offering an opportunity for host-specific selection to exploit segregating matrilineal variation (Fig. 1C and fig. S11). This pattern is reversed within *optatus*, where autosomal variation is less associated

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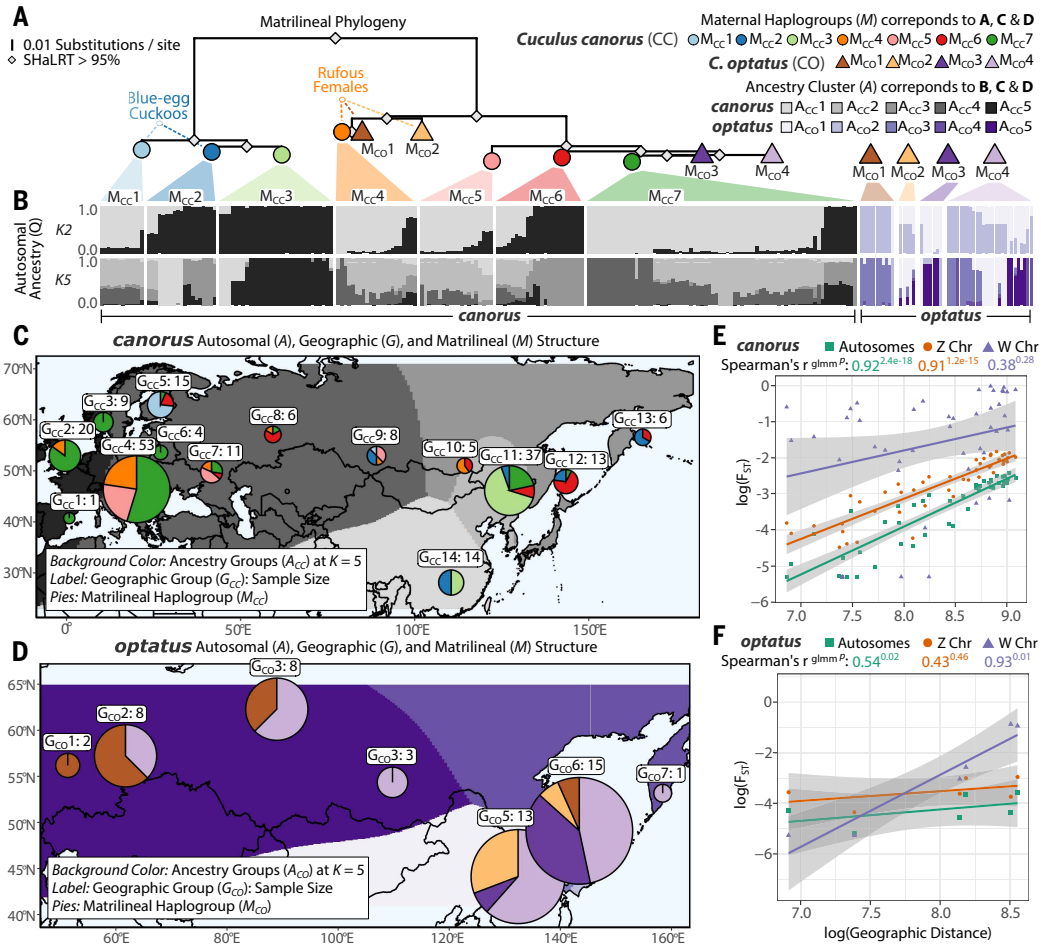


Fig. 1. Genetic structure across the geographic range of *C. canorus* and *C. optatus*. (A) Matrilineal phylogeny of the W chromosome ($N = 51,605$ SNPs) with truncated branches identifying seven *canorus* (CC) maternal haplogroups (M_{cc1} to M_{cc7}) and four *optatus* (CO) haplogroups (M_{co1} to M_{co4}). Diamonds indicate nodes with Shimodaira-Hasegawa approximate likelihood ratio test (SHaLRT) support $>95\%$. (B) Ancestry analysis ($n = 202$ *canorus*, 50 *optatus*) inferred from unlinked autosomal SNPs ($N = 1,546,013$ *canorus*, $920,854$ *optatus*), arranged by maternal haplogroup. (C) Spatial structuring of *canorus*. Background color (grayscale; fig. S6) indicates the spatial distribution of autosomal ancestry (A_{cc1} to A_{cc5}) inferred using ADMIXTURE at (K) = 5. Pie charts indicate maternal haplogroups, with size corresponding to sample size. Samples are clustered into 14 geographic groups (G_{cc1} to G_{cc14}) by k-means on latitude and longitude; group labels and sizes are shown above the pie charts. Please see supplementary materials for alternative symbology (figs. S8 and S11). (D) Spatial structuring of *optatus*. (E) Correlations between geographic distance (km) and genetic distance in *canorus*, summarized from autosomal (F_{ST}) or matrilineal (Φ_{ST}) variation across pairwise comparisons between geographic groups. Spearman's rank correlation (r_s) and generalized linear mixed model (glmm) Bonferroni-adjusted p (shown in superscript) for each genomic compartment are indicated above, with shading indicating 95% confidence intervals. (F) Correlations for *optatus*.

with geographic distance (Spearman's $\rho = 0.54$, $p = 1.0$) than matrilineal variation (W chromosome $\rho = 0.93$, $p = 0.03$) (Fig. 1, D and F, and fig. S10).

Genomic correlates of egg diversity

Next, limiting the dataset to females with known maternal egg phenotype (E_{cc} : $n = 60$ *canorus*; E_{co} : $n = 27$ *optatus*), we assessed whether *canorus* and *optatus* egg diversification follows strict matrilineal inheritance (5, 19, 21, 28) (M_{cc} , M_{co}) or is better explained by autosomal contributions (24, 25) (A_{cc} , A_{co}) or geographic location reflecting local host availability (G_{cc} , G_{co}) (Fig. 2, A and B, and fig. S12). We replicated the following analyses with an expanded dataset that included males ($n = 86$ *canorus*, 50 *optatus*), which yielded qualitatively identical patterns despite restricting the matrilineal component to mitochondria (figs. S13 and S14).

Within *optatus*, where egg phenotype variation is limited to differences in maculation on shared eggshell coloration and egg size (29), geographic location is the primary predictor (distance-based redundancy

analysis pseudo-adjusted $R^2 = 0.78$), with weaker associations for matrilineal structure ($R^2 = 0.04$) and autosomal variation ($R^2 = 0.02$) (Fig. 2C). This finding is supported by classical machine learning, as adding a matrilineal component to geographic and autosomal predictors did not improve model predictions (Fig. 2F and fig. S13). By contrast, variation in *canorus* egg morphs, differing in both eggshell coloration and maculation, are most strongly associated with matrilineal variation (distance-based redundancy analysis pseudo-adjusted $R^2 = 0.26$), followed by geographic location ($R^2 = 0.16$) and autosomal ancestry ($R^2 = 0.02$) (Fig. 2, A and C, and fig. S14). Matrilineal inheritance is further supported by kinship analyses of 107 related *canorus* individuals: Phenotype concordance is complete among maternal relatives (100% across all kinship levels), whereas paternal relatives, particularly diagnostic cousins, show significantly less concordance (75.8%; $n = 50/66$ matches; Fisher's exact test $p = 0.01$) (Fig. 2D and fig. S15). Machine learning results reinforce the primacy of matrilineal inheritance, as adding haplogroup information substantially improves classification of *canorus* egg morphs (Fig. 2E). The effect persisted, though diminished,

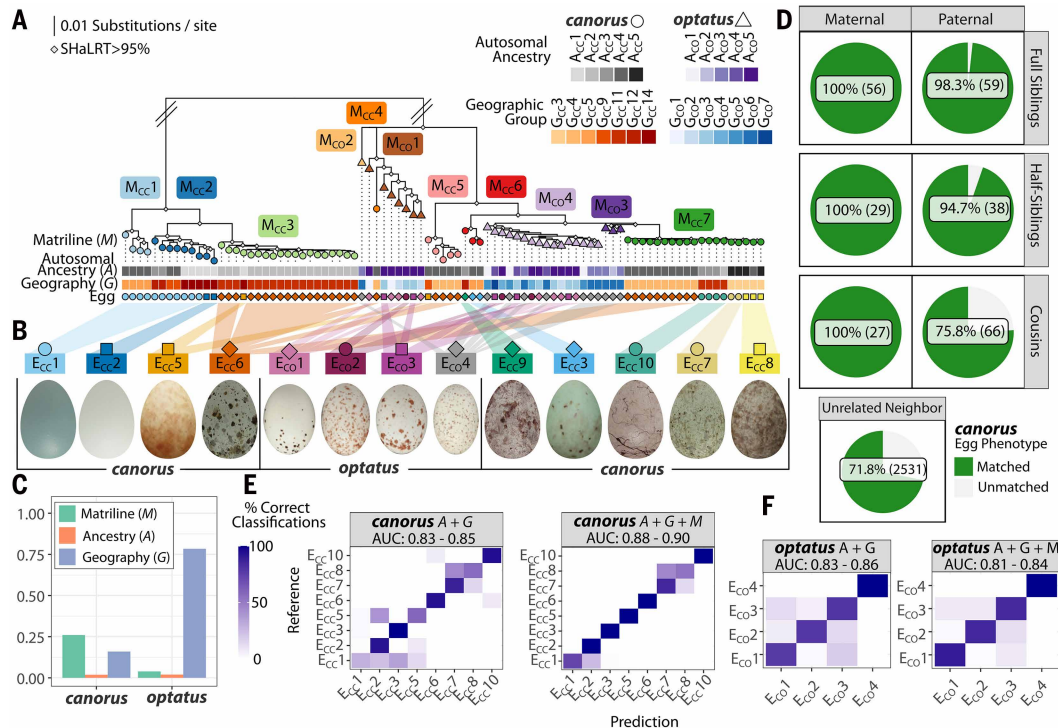


Fig. 2. Matrilineal variation is associated with egg diversification in *C. canorus*. (A) Matrilineal W chromosome phylogeny ($n = 60$ *canorus*, 27 *optatus*), with corresponding autosomal groups (A), geographic groups (G), and egg morphs indicated below each tip, each representing an unrelated female with known egg morph. Diamonds indicate node support exceeding 95% (SHaLRT). (B) Phenotypic variation within *canorus* (E_{cc}) and *optatus* (E_{co}) egg morphs. (C) Variation explained (pseudo- R^2) by continuous pairwise distance matrices of matrilineal, geographic, or autosomal variation estimated with a distance-based redundancy analysis (compare figs. S13 and S14 for replicated analyses with mtDNA variation). (D) Egg phenotype matching among varying degrees of related *canorus*. (E) Egg morph predicted with machine learning using either autosomal ancestry cluster (A) and geographic group (G) alone (left) or with the addition of haplogroup (M) (right). Confusion matrix of predictive modeling shows the proportion of correct classifications across 100 bootstrapped replicates. The area under the curve (AUC) 95% confidence interval across replicates is indicated below each label. (F) Confusion matrix for *optatus*.

when divergent M_{cc1} , M_{cc2} , and M_{cc3} haplogroups, including the immaculate blue egg (E_{cc1}) morph (19), were excluded (fig. S14G). However, subtle differences among most brownish egg morphs (E_{cc6} , E_{cc7} , E_{cc8}), which are largely associated with common *Acrocephalus* hosts, remain poorly resolved by any model (Fig. 2, B and E, and figs. S1 and S13). Stochastic character mapping of egg state on maternal and biparental phylogenies provided additional detail. By leveraging mitoautosomal discordance, it corroborated strong matrilineal association for both blue egg types E_{cc1} and E_{cc3} (figs. S16 to S18). Although E_{cc1} has previously been linked to a deeply divergent mitochondrial lineage associated with blue eggs [see (19)], greater parsimony of egg-state transitions on the mitochondrial tree for E_{cc3} suggests matrilineal association for a second, distinct blue-green egg phenotype (associated with brambling hosts; Fig. 2E and figs. S16 and S17).

Overall, these findings provide evidence for a matrilineal contribution of egg diversification in *canorus*, likely associated with changes in eggshell coloration evolving from a brownish, E_{cc6} -like ancestor (figs. S16 and S17), a widespread morph that allows exploitation of broadly distributed *Acrocephalus* hosts across the species range (28, 30–32). Male mate choice appears largely opportunistic and shaped by local host availability (3, 5, 25, 28, 33, 34), preventing stable genome-wide autosomal transmission through the patriline (8, 10) (Fig. 2, C and E). Egg patterning, which distinguishes many of the nonancient haplogroup morphs within *canorus* and accounts for basically all variation in *optatus* samples, also covaries with geographic structure, enabling adaptation to locally abundant and stable hosts through biparentally inherited genetic variation.

Genetic basis of egg phenotype

Next, we sought to identify candidate genes associated with egg phenotype. Under the assumption that genomic regions with elevated genetic differentiation may flag genetic variation related to egg morph (35), we performed an all-versus-one F_{ST} scan with resampling [bootstrapped F_{ST} (bF_{ST}) in 10-kb windows, 100-base pairs for mtDNA]. Consistent with the genome-wide analysis described earlier in the text, matrilineal elements (W chromosome, mtDNA) are enriched for the blue E_{cc1} morph, with W-linked enrichment relative to other egg morphs being more than twofold the permuted expectation (observed/permuted = 2.1, $p < 0.01$) (Fig. 3A and table S2). Conversely, significantly more autosomal outliers than expected were observed for E_{cc10} (enrichment = 1.9; $p < 0.01$). By contrast, *optatus* egg morphs show significant enrichment only on autosomes and the Z chromosome (Fig. 3A).

Genes in close proximity to the main outlier regions include credible functional candidates that influence the two main components of egg phenotype variation: eggshell coloration and maculation. Eggshell color variation, which is more similar between closely related females (36), results from the mixing of two heme-derived pigments: protoporphyrin IX (reddish-brown) and biliverdin (blue-green) (37). Maculation forms as the egg passes through the uterus and is shaped by the distribution of pigment glands, timing of pigment release, and mode of egg rotation (37, 38).

Of the biparentally inherited genes associated with differences in maculation, several relate to female reproductive function (table S3). In *canorus*, the top differentiated peak for E_{cc10} lies close to *RLN3*, which is involved in pituitary and oviduct signaling in birds (39). In

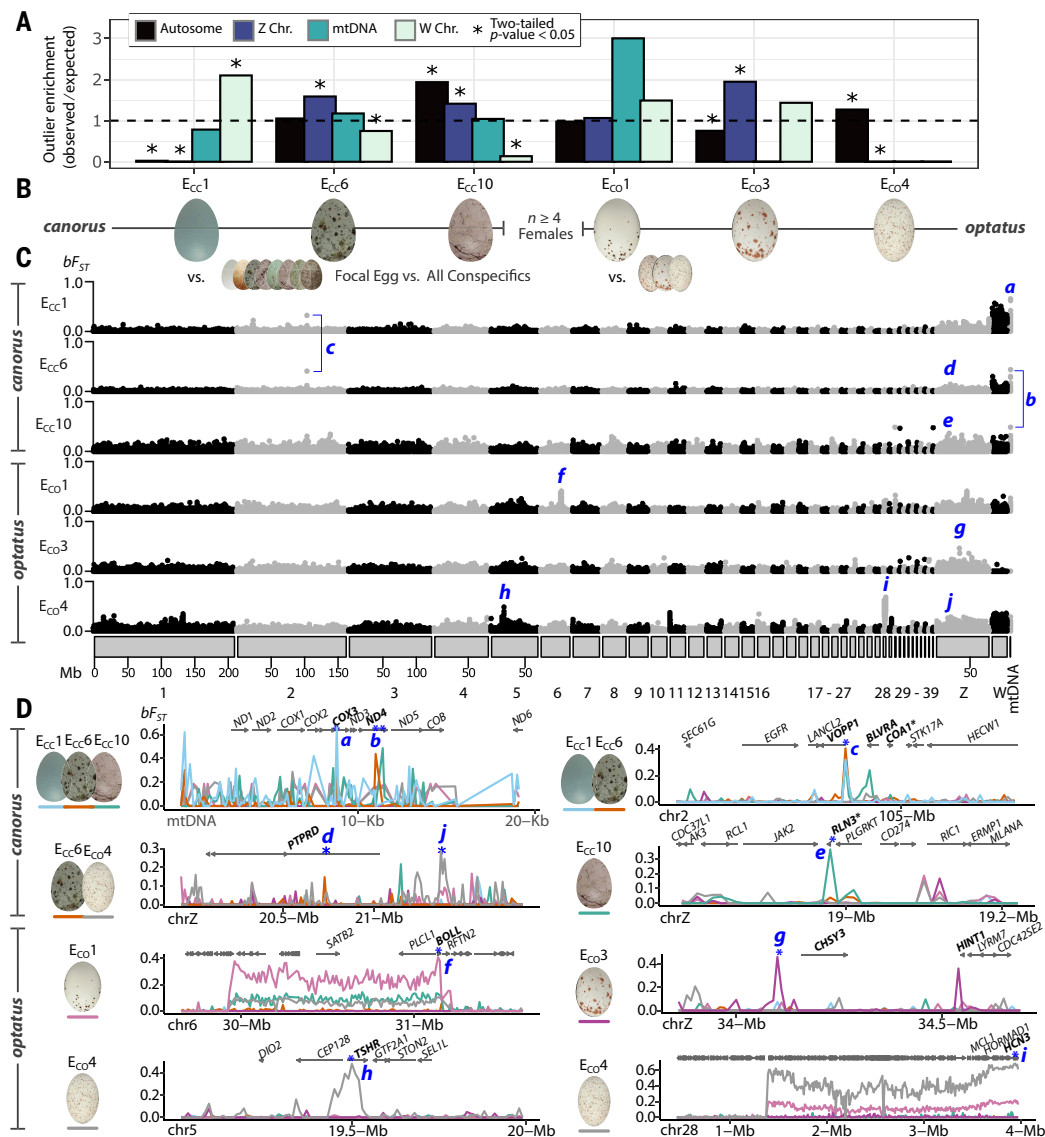


Fig. 3. Genetic basis of egg phenotype. (A) Permutation enrichment of bF_{ST} outliers per egg across chromosome class conducted separately for species. For joint analysis across species, see fig. S26 (Note: bF_{ST} is used throughout figure.) (B) Pairwise one-versus-all bF_{ST} comparisons, separated by species (*canorus* and *optatus*). (C) Genome-wide bF_{ST} in 10-kb windows, with the highest outlier windows indicated with blue letters. (D) bF_{ST} peak regions and nearby candidate genes (bold face), with the highest F_{ST} peak indicated with an asterisk. Line colors correspond to different egg morphs, with the primary candidate morph to the left of each plot. Black asterisks indicate unnamed genes with homology inferred from National Center for Biotechnology Information (NCBI) annotations (e.g., LOC104063035 = “relaxin-3”; LOC104065962 = “COA1-like”). The blue letters indicate the same outlier windows as in (C).

optatus, key outliers are linked to *BOLL* and *CHSY3* [involved in germ cell development (40, 41)], *HINT1* [linked to embryogenesis and sex hormones (42)], and *TSHR* and *HCN3* [involved in hormone signaling and eggshell traits (43, 44)] (table S3). Two regions show recombination suppression, potentially preserving trait combinations (fig. S19). *PTPRD*, linked to eggshell color in domestic chickens (45), is differentiated in both *Ecc6* (*canorus*) and *Eco4* (*optatus*) within ~500 kb of the same locus (Fig. 3D). Secondary peaks across egg morphs also implicate genes in female reproductive development (fig. S20).

In *canorus*, biparentally inherited genes linked to eggshell coloration are either directly involved in the heme biosynthetic pathway or in mitochondrial homeostasis, which regulates this process through metabolic signaling and shared biosynthetic resources (46–48). Together with studies implicating oxidative stress in female avian reproduction (49, 50), this suggests a tentative, mechanistic link of mitochondrial

function and eggshell coloration, which is also observed in chickens (51). Consistent with this hypothesis, candidate regions for *Ecc1* and *Ecc6* lie within 170 kb of *BLVRA*, a key gene in heme catabolism that reduces oxidative stress by converting biliverdin to bilirubin (52), as well as *VOPPI* and *COA1*, which are both likewise linked to mitochondrial function (53, 54) (Fig. 3, C and D). Analyses replicated using a relaxed threshold requiring three female nestlings for focal contrasts corroborated the association of *BLVRA* and *RLN3* and, for *Ecc8*, additionally implicated an autosomal gene involved in mitochondrial fission and associated with egg laying in chickens (55) (fig. S21 and table S3).

Isolating the matrilineal mutations for eggshell coloration is challenging owing to the large number of substitutions that have accumulated along long branches and the complete linkage of mitochondrial and W chromosome haplotypes. To identify candidate mutations, we

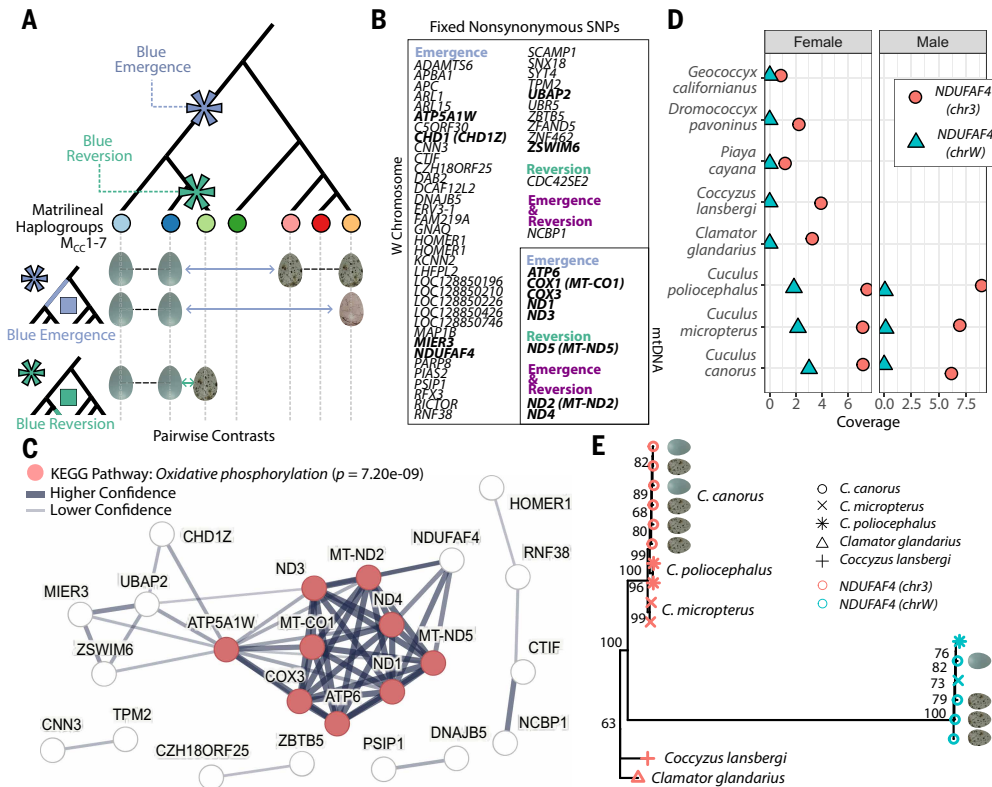


Fig. 4. *C. canorus* blue egg background predominantly maps to oxidative phosphorylation genes. (A) Schematic (top) depicting the evolutionary emergence and subsequent diversification of *canorus* egg morphs, with pivotal phenotypic shifts indicated by asterisks. Fixed differences in pairwise comparisons of haplogroups associated with same versus different egg types flag candidate mutations that contribute to phenotypic divergence. Investigated scenarios include the origin of the immaculate blue egg phenotype (blue emergence) and its subsequent reversal (blue reversion). Illustrative schematics (bottom) denote specific branches and mutations within the phylogeny where these evolutionary events are hypothesized to occur, along with the pairwise egg contrasts used to identify them. (B) Candidate genes containing a fixed nonsynonymous SNP for each contrast across the W chromosome and mtDNA. Genes in bold are shown in the Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway in (C). (C) Gene network associations among candidate genes. Note that all nonmitochondrial genes are found on the W chromosome (table S6). (D) Resequencing coverage of the autosomal and W chromosome paralog of *NDUFAF4* across Cuculiformes. (E) Maximum-likelihood phylogenetic tree of the *NDUFAF4* paralogs, with bootstrap support indicated at each node.

screened for fixed differences between pairs of matriline associated with an egg type (“matriline association mapping,” e.g., M_{CC}1:E_{CC}1 versus M_{CC}7:E_{CC}6) (Fig. 4A). We focused on the immaculate blue egg morph (E_{CC}1), which occurs across two maternal haplogroups (M_{CC}1 and M_{CC}2), persists thousands of kilometers apart, and yet remains at high frequencies in the respective populations, putatively owing to highly discriminatory hosts (Fig. 1C) (3, 56). Based on our matrilineal phylogenetic reconstruction, mutations along a long, ancient branch conferred the immaculate blue egg (blue emergence), followed by a reversion within this lineage to other eggshell colors in East Asia (blue reversion) (Fig. 4A). The remaining nonblue haplogroups (e.g., M_{CC}4 to M_{CC}7) diversified into other egg morphs but never to immaculate blue, based on present sampling and knowledge (Figs. 1A and 2A).

Consistent with the broad-stroke outlier analyses (Fig. 3), fixed single-nucleotide polymorphisms (SNPs) are predominantly observed in the matrilineal genome across both evolutionary scenarios. This is particularly pronounced for the origin of the blue egg (blue emergence), where 55% ($n = 123/225$) of mtDNA and 75% ($n = 33,597/44,739$) of W chromosomal SNPs are fixed, which is in stark contrast to none on the autosomes or Z chromosome (table S4). This pattern is reiterated for the reverted E_{CC}6 egg contrast (blue reversion) with 12% ($n = 9/73$) fixed

SNPs in mtDNA, 5% ($n = 413/8,490$) in the W chromosome, and none on the autosomes or Z chromosome. Mitochondrial genes *ND2* and *ND4* harbor nonsynonymous SNPs linked to both emergence and reversion of blue eggshell coloration (Fig. 4B), a pattern corroborated with de novo mitochondrial assemblies that further implicates *COX1* and *ND5* (figs. S22 and S23). Together with biparentally inherited genes associated with mitochondrial homeostasis (Fig. 3D), these fixed nonsynonymous mutations could be implicated in mitochondrial regulation for eggshell coloration.

Although a mitochondrial shift in egg phenotype mediated by the heme pathway is mechanistically plausible, several W-linked genes containing a fixed nonsynonymous SNP also interact with the statistically enriched oxidative phosphorylation pathway ($p = 7.20 \times 10^{-9}$) (Fig. 4C and table S5). Among them, an *NDUFAF4*-like gene, a homolog of *NDUFAF4* acting in the respiratory complex I assembly, represents a candidate for further functional investigation (LOC128850245 in RefSeq Assembly GCF_019796375.1; table S6). It has a copy on the W chromosome (28 kb; mean female coverage = 6.4×, male coverage = 0.13×) in addition to an ancestral autosomal copy on chromosome 3 (4.0 kb; female coverage = 11×, male = 11×), which exhibits no autosomal differentiation among *canorus* egg morphs (Fig. 4D and fig. S24). Phylogenetic analysis across the Cuculiformes order suggests that the W paralog was acquired at the onset of the brood parasitic genus *Cuculus* (Fig. 4E and table S6).

The W-linked copy contains the same number of exons and has full open reading frames (fig. S25) yet is substantially diverged from its autosomal paralog since insertion in the *Cuculus* ancestor (Fig. 4E). Although mitonuclear cross-talk has broad biological implications (57), our results suggest the possibility that polygenic autosomal-mitochondrial-W chromosome interactions contribute to egg background color diversification. Other avian species have co-opted autosomal genes to produce immaculate blue eggs, including SNPs upstream of efflux transporter *ABCG2* in ducks (58), an endogenous retroviral insertion upstream of anion transporter *SLCO1B3* in chickens (59), and a recessive gene in quail (60), illustrating the multiple mutational pathways that underlie egg diversification in birds.

Conclusions

In *canorus*, phenotypic variation in egg traits, especially eggshell coloration, appears primarily mediated by matrilineally inherited genes, as suggested nearly a century ago (6). This remains surprising, as exclusive matrilineal inheritance poses great challenges: Reduced effective population size limits the efficacy of selection, and beneficial mutations compete among matrilineal rather than recombining into advantageous trait combinations. Thus, barring new mutations, we would not expect, for example, a female with rufous plumage (associated with

haplogroup M_{CC4}) to lay immaculate blue eggs (associated with M_{CC1} and M_{CC2}), supporting earlier speculations by Kunz that plumage coloration is linked to host choice and egg coloration (61). Additionally, the limited mutational target on mtDNA and the W chromosome may require recruitment of autosomal genes, as suggested here for the translocation of *NDUF4F4*. Despite these constraints, female-biased gene flow involving multiple matrilineal lines maintained by balancing selection (23) appears to provide sufficient adaptive variation for exploiting multiple host species.

Biparental genetic contributions and local geographic structure also seem to contribute to phenotypic diversity. In *canorus*, co-option of nuclear genes with mitochondrial-related functions suggests that interactions between matrilineal and autosomal variation may contribute to eggshell coloration. By contrast, in *optatus*, variation is limited to maculation with exclusive contribution of autosomal genes. Such polygenic adaptation—which is also observed in egg phenotype variation of the village weaver (*Ploceus cucullatus*), a host species of Diederik cuckoo that exhibits extensive intra- and interclutch diversity as a defense against parasitism (62)—may develop in *canorus* over prolonged coexistence with abundant, accessible host species, such as *Acrocephalus* warblers (28, 30–32). Although selective pressures differ for hosts and parasites, the involvement of many unlinked autosomal loci in both systems may reflect convergent selection on egg phenotype diversity.

Host race diversification in the common cuckoo seems largely predicated on matrilineal variation, limiting the potential for speciation. Though biparentally inherited loci contribute to regional egg diversity, mostly for maculation in *optatus*, they fail to establish strong, genome-wide barriers to gene flow that would place gentes on independent evolutionary trajectories.

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SUPPLEMENTARY MATERIALS

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Materials and Methods; Tables S1 to S7; Figs. S1 to S32; References (64–126); MDAR Reproducibility Checklist

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